

Four Routes Through Hyperspace: Wavetable Navigation Strategies via Givens Rotations

Luca Bimbi^{1,2}, Anthony Di Furia^{3,4}, and Giuseppe Silvi⁵ *

¹ Conservatorio “A. Corelli”, Messina, Italy

² Conservatorio “S. Giacomantonio”, Cosenza, Italy

`bimbi.luca@gmail.com`

³ LEAP – Laboratorio ElettroAcustico Permanente, Rome, Italy

⁴ Conservatorio “N. Piccinni”, Bari, Italy

`anthonydifuria.sound@gmail.com`

⁵ Conservatorio “A. Casella”, L’Aquila, Italy

`grammaton@me.com`

Abstract. This paper presents an approach to navigating the hyperspace of a wavetable, interpreted as a vector in $\mathbb{R}^N$, based on Givens rotations. A discrete audio signal of N samples is treated as a point in N -dimensional Euclidean space; Givens rotations, acting on pairs of coordinates through the sine and cosine of an angle, displace this point while preserving the L^2 norm. Four strategies for selecting rotation pairs and angles are proposed and compared: a local adjacent chain, a non-local prime step, Sobol quasi-random sequences, and a Kuramoto coupled-oscillator model. Each strategy is realized both as an interactive Python prototype and as a Csound plugin opcode operating on 1024-sample wavetables. A preliminary, qualitative audio observation of the four strategies under controlled parameter sweeps is also reported, with a supplementary rendering dataset.

Keywords: wavetable synthesis, Givens rotations, hyperspace navigation, Csound opcodes, Kuramoto model, Sobol sequences

1 Introduction

Wavetable synthesis stores a single waveform cycle in a table of N samples read cyclically at a variable rate [18]. Each wavetable is an ordered sequence of N numbers – a point in $\mathbb{R}^N$. Shannon’s geometric representation of a signal as a point in N -dimensional space [19] supports this interpretation: the L^2 norm $\|\mathbf{x}\|_2$ encodes total signal energy, and the group $O(N)$ of orthogonal transformations preserves it [9]. Traditional interpolation between predefined wavetables constrains navigation to linear trajectories between known points. The approach proposed here departs from a single initial wavetable and reaches a large set of waveforms through elementary orthogonal transformations.

Givens rotations [8] – two-component orthogonal transformations – serve here as the building block for direct, real-time waveform-space navigation, without the training datasets required by neural-network approaches [6,7]. Section 7 complements this geometric picture with a preliminary, qualitative observation of the four strategies’ audio output.

2 Signals as Points in a Hyperspace

A discrete audio signal $\mathbf{x} = (x_0, \dots, x_{N-1}) \in \mathbb{R}^N$ lies on a hypersphere whose radius is the norm $\|\mathbf{x}\|_2$ [19,11,24]. The rotation group $SO(N)$ contains every norm-preserving proper rotation; its dimension $N(N-1)/2 = 523\,776$ for $N = 1024$ counts the independent rotation planes available, not the dimensionality of the reachable set itself, which is the hypersphere S^{N-1} [9,22]. Modifying any sample displaces the point along this hypersphere. Navigating the space therefore amounts to choosing, at each step, the pair of samples that defines the rotation plane and the angle through which they exchange energy.

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3 Givens Rotations

A Givens rotation $G(i, j, \theta)$ acts on two components only, leaving all others unchanged [8,9]:

$$x'_i = \cos \theta \cdot x_i - \sin \theta \cdot x_j, \quad x'_j = \sin \theta \cdot x_i + \cos \theta \cdot x_j \quad (1)$$

The quantity $x_i^2 + x_j^2$ is invariant, guaranteeing energy preservation, while x_k ($k \neq i, j$) is untouched. Any rotation in $\mathbb{R}^N$ decomposes into a sequence of Givens rotations [9]. Such rotations also underlie QRD-RLS adaptive filtering [15,14,2], though there for numerically stable matrix factorization rather than waveform navigation. The choice of pair (i, j) and angle θ at each step determines the trajectory through waveform space.

3.1 Example in $\mathbb{R}^3$

Starting from $\mathbf{x} = (1, 1, 1)$, $\|\mathbf{x}\|_2 = \sqrt{3}$, two successive rotations illustrate energy transfer: $G(0, 1, \pi/4)$ yields $\mathbf{x}' = (0, \sqrt{2}, 1)$; $G(1, 2, \pi/4)$ yields $\mathbf{x}'' \approx (0, 0.293, 1.707)$. Both lie on the sphere of radius $\sqrt{3}$; energy has migrated from x_0 to x_2 (Fig. 1).

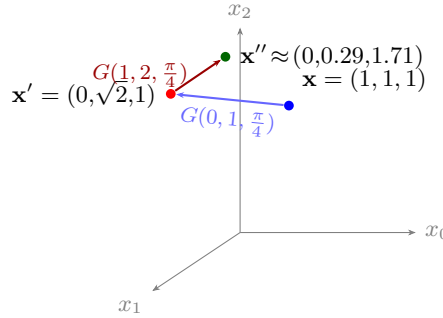


Fig. 1. Two Givens rotations in $\mathbb{R}^3$: the point moves along the sphere of radius $\sqrt{3}$, transferring energy from x_0 to x_2 .

3.2 Properties of Composition

Every rotation is invertible by $G(i, j, -\theta)$. The order of rotations matters when pairs share an index [9]. Composition preserves L^2 norm but not the signal peak; a peak normalization ($\max |x_k| = 1$) is therefore applied after each rotation sequence. The rescaling itself is not norm-preserving: it projects the point onto a hypersphere of different radius, so the energy invariance holds for the rotation sequence alone. In practice the normalization is indispensable for real-time amplitude control, and the geometric model should be read as describing Givens navigation on the original hypersphere, from which the output signal is drawn up to a global amplitude factor.

4 Four Rotation Strategies

All four strategies apply Givens rotations to a working copy of the wavetable, then peak-normalize. The original ftable is never modified. In the three deterministic strategies angles decay geometrically: $\theta_t = \theta_0 \cdot kmove \cdot decay^t$; $kmove = 0$ returns the unmodified table.

4.1 Local Adjacent Rotations (wtgivlocal)

Pairs of adjacent samples $(k, k+1)$ are rotated starting from a pivot *start*. Three chains are available: *right* (toward $N-1$), *left* (toward 0), and *both* (bidirectional expansion). The both-chain produces a bell-shaped perturbation centred on *start*; geometric decay ensures distant samples are minimally affected.

4.2 Non-Local Prime-Step Rotations (wtgivnlocal)

Pairs $(i, (i + p) \bmod N)$ with $p = 37$ (coprime with 1024 [10]) connect distant samples; the sequence visits all N positions before repeating. Rotation counts per direction: $\lfloor d \cdot 512 \rfloor$, $d \in \{\text{right}, \text{left}, \text{both}\}$. From the first step, energy is redistributed globally, producing more radical displacements than the local strategy.

4.3 Sobol Quasi-Random Angles (wtgivsobol)

The previous strategies push angles in a constant direction. The one-dimensional Sobol sequence [21,16] generates low-discrepancy values in $[0, 1)$ that cover the interval more uniformly than pseudo-random numbers, alternating sign quasi-uniformly. The update rule uses a 32-bit accumulator x_n :

$$x_n = x_{n-1} \oplus v_c, \quad c = \text{ctz}(n), \quad v_c = 2^{31-c} \quad (2)$$

Each value $u_t = x_n/2^{32}$ is mapped to $\theta_t = \theta_0 \cdot \text{kmov} \cdot \text{decay}^t \cdot (2u_t - 1)$. Pairs follow the non-local prime-step scheme. The Sobol generator resets at every k-block, ensuring full determinism.

4.4 Kuramoto Coupled Oscillators (wtgivkura)

The Kuramoto strategy [12,23] introduces *time*: rotation angles are phases of a dynamical system that evolves continuously across k-blocks. The model governs $K_{\text{dim}} = 64$ oscillators:

$$\frac{d\vartheta_i}{dt} = \omega_i + \frac{K}{K_{\text{dim}}} \sum_{j=1}^{K_{\text{dim}}} \sin(\vartheta_j - \vartheta_i) \quad (3)$$

Natural frequencies $\omega_i = \theta_0 \cdot \text{decay}^i \cdot (1 + \text{tilt} \cdot p_i)$ with $p_i \in [-1, 1]$ can be biased via *right* and *left*. Integration uses RK2 [5] with a fixed step $dt = 0.002 \cdot \text{kmov}$, so the speed of the trajectory scales with the control rate; anchoring the step to the control period instead, $dt = (\text{kmps}/\text{sr}) \cdot \text{kmov}$, would in principle make it independent of *kmps*. When coupling K (parameter *kboth*) exceeds the synchronization threshold [23,1], phases lock and the wavetable converges; below threshold, phases diverge and the waveform evolves continuously. Phase persistence across k-blocks gives the system memory of its trajectory.

5 Python Implementation and Comparison

The reference implementation is the set of Csound opcodes of Section 6; an interactive `matplotlib` prototype served as design and verification environment, and the comparisons below are its output ($N = 8$, base sinusoid $x_n = \sin(2\pi n/8)$). A Csound-native visualization is possible by plotting base and transformed tables with `display`.

With identical parameters ($\text{start} = 3$, $\text{both} = 1.0$, $\theta_0 = 0.785$, $\text{decay} = 1.0$, $\text{kmov} = 1.0$) the three deterministic strategies reach distinct points in $\mathbb{R}^8$ (Table 1, Fig. 2).

Table 1. Normalized sample values for the three deterministic strategies ($\text{start} = 3$, $\text{both} = 1.0$, $\theta_0 = 0.785$, $\text{decay} = 1.0$, $\text{kmov} = 1.0$).

k	0	1	2	3	4	5	6	7
Base	0.000	0.707	1.000	0.707	0.000	-0.707	-1.000	-0.707
Local	-0.161	0.173	0.697	1.000	0.793	0.561	-0.077	-1.000
Non-loc.	-0.197	-1.000	-0.948	-0.262	0.167	0.651	0.519	0.269
Sobol	-0.107	0.901	-1.000	0.819	-0.194	0.246	-0.107	-0.185

The local strategy preserves the overall waveform shape and concentrates energy near $\text{start} = 3$, while the non-local one inverts samples 1–3, a far larger displacement in $\mathbb{R}^8$; the Sobol strategy produces rapid sign alternations. The parameter *decay* controls trajectory depth: 1.0 gives uniform rotation; smaller values concentrate the perturbation near *start*.

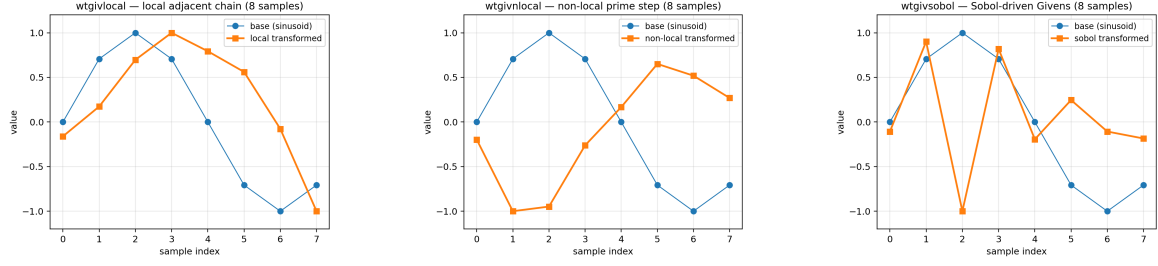


Fig. 2. Left to right: local (`wtgivlocal`), non-local (`wtgivnlocal`), Sobol (`wtgivsobel`). Blue: base sinusoid; orange: transformed wavetable ($N = 8$).

6 Csound Implementation

The four strategies are realized as plugin opcodes in C [13,4], operating on $N = 1024$ samples ($p = 37$, $MAX_STEPS = 512$, $K_{\text{dim}} = 64$) [3]. All share the syntax:

```
aOut opcode kfreq, kamp, itab, kstart, kright,
      kleft, kboth, ktheta0, kdecay, kmove
```

At each k-block: (1) copy wavetable; (2) apply rotations; (3) peak-normalize; (4) generate audio by linear interpolation. Multiple instances may share the same ftable. Interpolation concerns only the read-out stage: the geometric model decides which point of the table is reached, the interpolator how the signal is read from it. The triangular kernel preserves neither band limitation nor energy, as ideal reconstruction would require sinc interpolation, but its low cost makes it the standard choice in real time [20].

Table 2 summarizes parameter semantics; Table 3 gives worst-case rotation counts per k-block. All opcodes run in real time at k-rate ≈ 689 Hz ($f_s = 44100$ Hz, $ksmps = 64$); the dominant cost, trigonometric computation, is reducible via lookup tables [20].

Table 2. Control parameter semantics.

Param.	Loc./NLoc./Sobol	Kuramoto
kstart	starting index	starting index
kright	right density [0..1]	ω bias R [0..1]
kleft	left density [0..1]	ω bias L [0..1]
kboth	bidir. density [0..1]	coupling K [0..10]
ktheta0	initial angle [rad]	ω_i scale [rad]
kdecay	geom. decay [0..1]	ω decay [0..1]
kmove	intensity [0..1]	speed dt [0..10]

Table 3. Cost per k-block (worst case).

Opcode	Max rot.	Notes
wtgivlocal	1023	bidir. chain
wtgivnlocal	1536	3×512
wtgivsobel	1536	+ Sobol gen.
wtgivkura	64	$+O(K_{\text{dim}}^2)$ RK2

A minimal Csound example:

```
giSin ftgen 0, 0, 1024, 10, 1
instr 1
  aOut wtgivlocal 220, 0.5, giSin, 3, 0.0, 0.0,
    1.0, 0.785, 0.9, line:k(0, p3, 1)
  outs aOut, aOut
endin
```

`kmove` ramps from 0 to 1 over the note, morphing the sinusoid into the rotated wavetable.

7 Discussion and Conclusions

Determinism and Temporal Behaviour. The local, non-local, and Sobol strategies are fully deterministic: identical parameters always yield the same wavetable. The Kuramoto strategy is history-dependent; the synchronization transition, controllable via K , adds a temporal dimension absent from the static strategies. Phases can be reset at note onset (`op_init()`) for reproducibility [3].

Preliminary Spectral Observations. Single-parameter sweeps were rendered for each opcode at a fixed carrier frequency (100 Hz) and inspected via spectrograms. Within the limits of this preliminary, non-exhaustive set of renderings, the local, non-local and Sobol strategies appear to return to the base sinusoid once `kmove` returns to its resting value, consistent with their deterministic formulation;

the Kuramoto strategy’s output did not appear to show the same return within the tested reset window, consistent with its phase-persistence. A slow sweep of the coupling parameter K (`kboth`) also appeared to show a spectral transition near where a synchronization threshold would be expected, though systematically characterizing it falls outside the scope of the present sweeps. Audio renderings, spectrograms, and parameter schedules are available as supplementary material.⁶

Geometry of the Strategies. The three deterministic strategies reach markedly different points in $\mathbb{R}^N$ from the same starting point and parameters (inter-strategy distances ≈ 1.5 – 2.5 against a hypersphere radius of 2.0). Whether distinct strategies can be made to converge toward nearby regions, which would point to attractor-like zones in the waveform hyperspace, remains an open question, and answering it will require metrics in both the sample and spectral domains [17].

Four strategies – local, non-local, Sobol, Kuramoto – have been formalized and implemented as Csound plugin opcodes; source code is freely available [3]. The preliminary spectral observations above appear consistent with the deterministic/history-dependent distinction between strategies; a broader evaluation with quantitative spectral metrics and listening tests is left to future work.

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⁶ Dataset available at <https://doi.org/10.5281/zenodo.22539741>, with a couple of figures also mirrored in the code repository [3].